

Homework 1 - Handwritten Solution Worksheet

EEE 554 - Probability and Random Processes | Fall 2026

Write your name: Jon Nethercraf Due: Friday, August 28 on Canvas

This worksheet reproduces the assigned questions and reserves space for your own handwritten derivations. Show the event definitions, formulas, and reasoning used for each answer.

1. Uniform reboot time (5 points)

A computer is programmed to reboot automatically for updates between 4:10 am and 4:55 am. The computer is equally likely to reboot anytime within this time interval.

- What is the probability that the computer will reboot at exactly 4:16 am?
- What is the probability that the computer will reboot between 4:15 am and 4:30 am?

a) If $T = t$ A continuous RV So probability is zero

b) If $4:15 - 4:30$ is an interval $5 \leq T \leq 20$
 $\frac{15}{45} = \frac{1}{3}$

2. Events in a two-dimensional sample space (15 points)

The sample space $S = \{(x, y): 0 \leq x \leq 3, 0 \leq y \leq 3\}$ includes values of x, y in the two-dimensional Cartesian space. Events are $A = \{(x, y): y \leq 3 - x\}$ and $B = \{(x, y): y \geq x\}$.

- Draw the regions corresponding to events A and B in the (x, y) -plane.
- Evaluate $P(A)$, $P(B)$, and $P(A | B)$.
- Are A and B independent? Justify your answer.



b)

$$A = 1 \times 3 \cdot 3 = 9/2, \quad P(A) = \frac{9/2}{9} = \frac{1}{2}$$

$$P(A|B) = \frac{(1/4)(1/2)}{1/2} = \frac{1}{4}$$

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$$y \leq 3; \quad P(A \cap B) = 1/4 = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = P(A)P(B)$$

$$P(A|B) = P(A) = \frac{1}{2}$$

3. Discrete tardiness distribution (10 points)

A habitually tardy person arrives at the theater late by s_i minutes, where $s_i = i$ for $i = 1, 2, 3, \dots$. If $P[s_i] = (1/2)^i$, what is the probability that they will be more than one minute late?

$$P(s_1, s_2) = P(s_2) + P(s_3) + \dots = \sum_{i=2}^{\infty} (1/2)^i = 1/4 \left(1 - 1/2\right) = 1/8$$

4. Drawing with replacement (10 points)

An urn contains 3 red balls and 2 black balls. Two balls are chosen in succession. The first ball is returned to the urn before the second ball is chosen. Each ball is chosen at random.

(a) What is the probability of choosing first a red ball and then a black ball?

(b) More generally, if the urn has k red balls and $N - k$ black balls, what is the probability of a red ball followed by a black ball when two balls are chosen at random with replacement?

Probable Ball is replaced

a.) $P(\text{Red then Black}) = P(\text{Red}) P(\text{Black}) = (3/5) (2/5) = 6/25$

b.) $P(R \text{ then } B) = 1/5 (N-1)/N^2$

5. Shared birthday (10 points)

A probability class has N students enrolled. What is the probability that at least two of the students will have the same birthday?

$$P(A_1) = (363/365) (364/365) \dots (365 - N + 1) / 365$$

for $N \approx 365$

for $N \gg 365, P \approx 1$

6. Independence with a fair die (10 points)

Given a fair die, consider A = getting 2 or 3 and B = getting an even number. Are A and B independent? (Hint: compare $P[A]$ and $P[A | B]$.)

$$A = 2, 3, \quad 50 \quad P(A) = \frac{2}{6} = \frac{1}{3} \quad ; \quad B = 2, 4, 6$$

$$P(A|B) = \frac{1}{3}$$

A & B are independent

7. Conditional events with a fair die (10 points)

Toss a fair die and consider A = getting an even number, B = getting a number greater than 2, and C = getting a number less than 5. Find $P[C]$, $P[B | C]$, $P[A | B, C]$, and $P[ABC]$.

$$C = 1, 2, 3, 4 \quad P(C) = \frac{2}{3}$$

$$B \text{ and } C \text{ are } P(B|C) = \frac{2}{4} = \frac{1}{2}$$

$$P(A|B, C) = \frac{1}{2}$$

$$A \text{ and } B \text{ and } C = 4 \quad \text{so} \quad P(ABC) = \frac{1}{6}$$

8. Cancer-test Bayes calculation (10 points)

Suppose 0.001% of the general population has a certain cancer. A blood test is positive with probability 0.99 if a patient has the cancer. It produces a false positive with probability 0.2 if the patient does not have cancer. If the test comes back positive, what is the probability that the person has cancer?

$$P(C) = .001\% = .00001$$

$$P(+|C) = .99 \quad \text{and} \quad P(+|C^c) = .2$$

$$P(C^+) = (.99)(.00001) / (.99)(.00001) + (.2)(.99999) =$$

$$.00004950$$

$$P = .000495\% \quad \text{or} \quad \text{about } 1 \text{ in } 2005$$

9. Repeated fax dialing (10 points)

A fax machine dials a phone number that is typically busy 80% of the time. It dials every 5 minutes until the line is clear and the fax can be transmitted. What is the probability that it will have to dial the number 9 times?

$$\begin{aligned} P(\text{busy}) &= .8 & a_2 &= .2 & P(\text{clear}) &= .2 \\ P(K=9) &= .8^8(.2) & & & & \\ &= .3,36\% \end{aligned}$$

10. Forming the word committee (10 points)

A person chooses 9 letters at random from the English alphabet with replacement. What is the probability that they will be able to make the word committee?

$$\begin{aligned} a_1 \cdot (a_1 \cdot a_1 \cdot a_1) &= 45,360 \\ \text{or } 1/26^9 & \cdot P = a_1 \cdot (a_1 \cdot a_1 \cdot a_1) \cdot (1/26)^7 \\ &= 45,360 / 26^9 \quad 8.75 \times 10^{-5} \end{aligned}$$